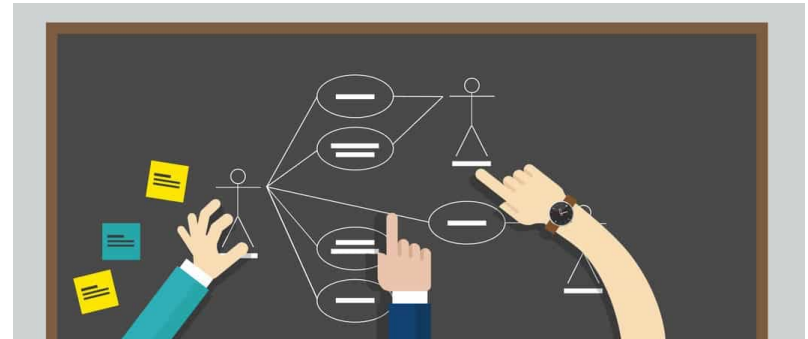


Chapter 7:

Use Case Diagram - 2



Recap: Use case diagram elements and notations

System boundary

- Sets a system's scope to use cases.

Actor

- User that interacts with a system.

Use case

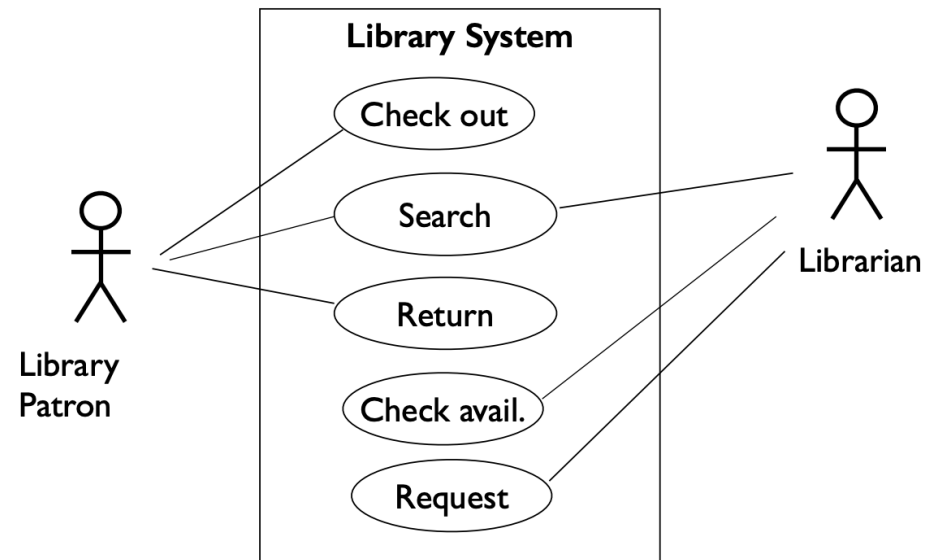
- Written description of the user's interaction with the software product to accomplish a goal.

Relationship

- Association is the relationship between an actor and a use case.
- Generalisation of an actor
- Generalisation of a use case
- Extend between two use cases
- Include between two use cases

Example #1 Library System

- Since both patron and librarian can initiate the use of the Library System, they are both primary actors.
- Since there are more than one primary actors, Librarian is placed on right side of the system.



Example #2 Inventory System

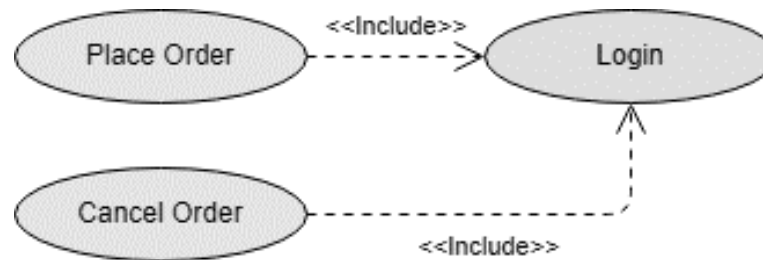
More complicated relationships

- Include between two use cases
- Extend between two use cases
- Generalisation of a use case
- Generalisation of an actor

Include Relationship between two Use Cases

*When a machine makes coffee, it **must** take in water.*

*When a user needs to place order, he **must** login first.*



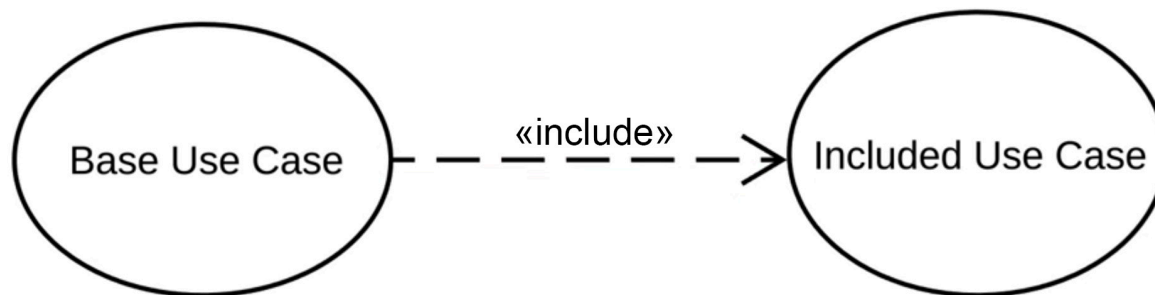
Include Relationship between two Use Cases

Place included use cases to the right of the base use case.

- This is done to improve readability and add clarity.

Draw a dashed arrow with an open arrowhead from the base use case to the included (common part) use case.

The arrow is labelled with the keyword «include».

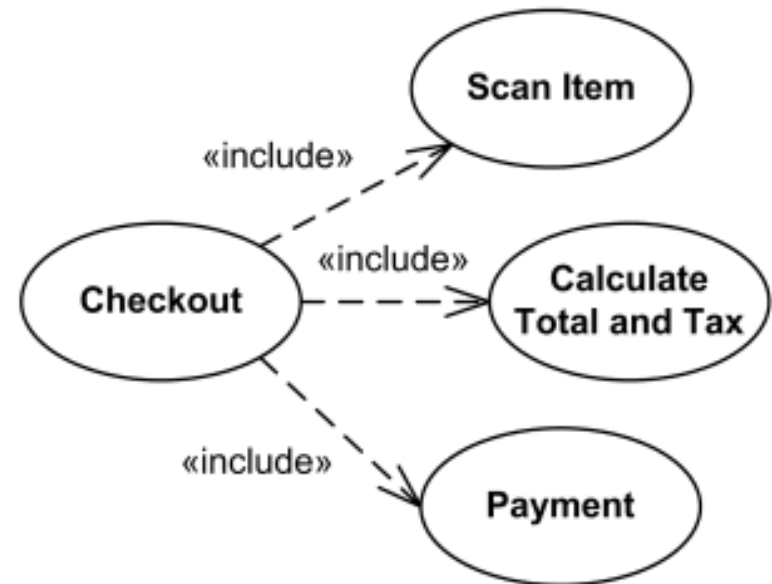


Include Relationship between two Use Cases

Checkout use case includes several use cases

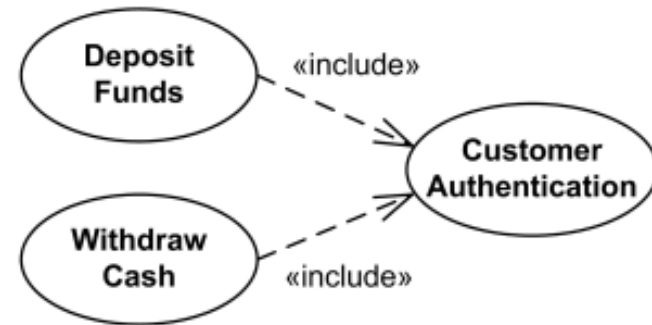
- Scan Item
- Calculate Total and Tax
- Payment

Checkout use case becomes incomplete by itself and requires included use cases to be complete.



Include Relationship between two Use Cases

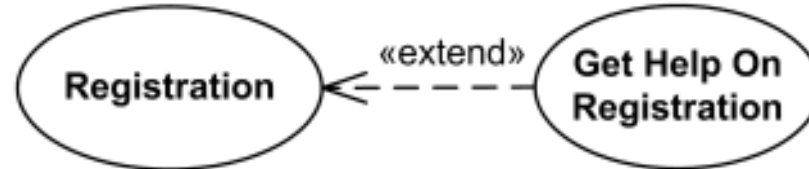
Deposit Funds and Withdraw Cash use cases include Customer Authentication use case.



Extend Relationship between two Use Cases

Extending Use Case defines optional behaviour.

When I order coffee, sometimes I order sugar and milk if I feel like it.



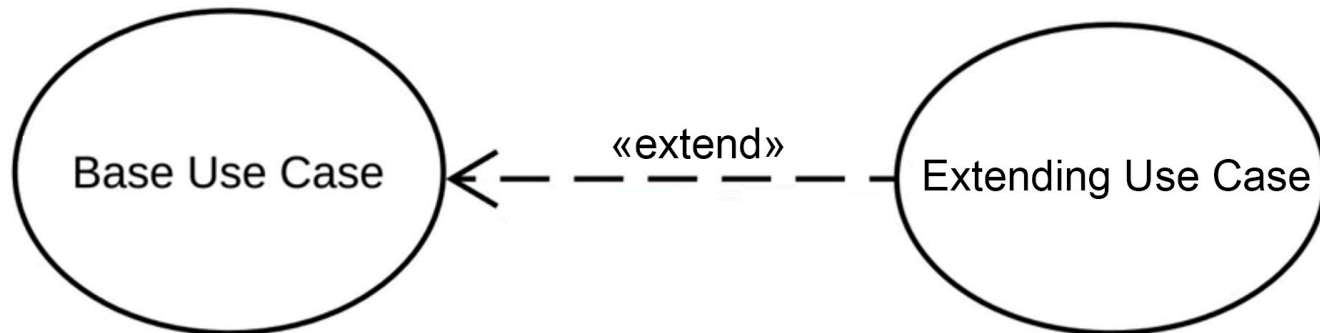
Registration use case is complete and meaningful on its own.

It could be extended with optional Get Help On Registration use case.

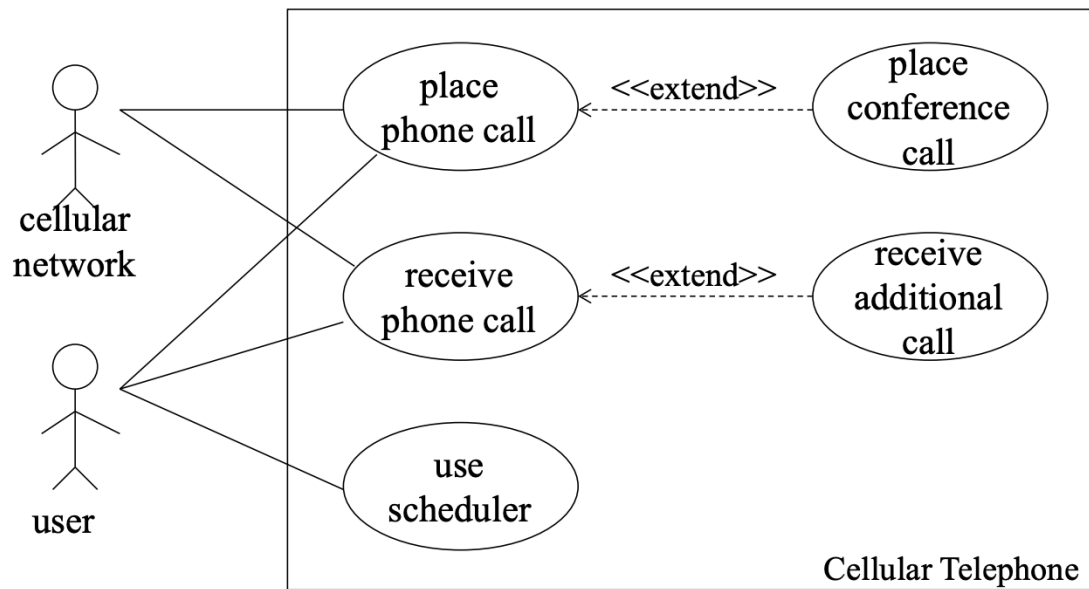
Extend Relationship between two Use Cases

Extend relationship is shown as a dashed line with an open arrowhead directed from the extending use case to the base use case.

The arrow is labelled with the keyword «extend».



Extend Relationship between two Use Cases



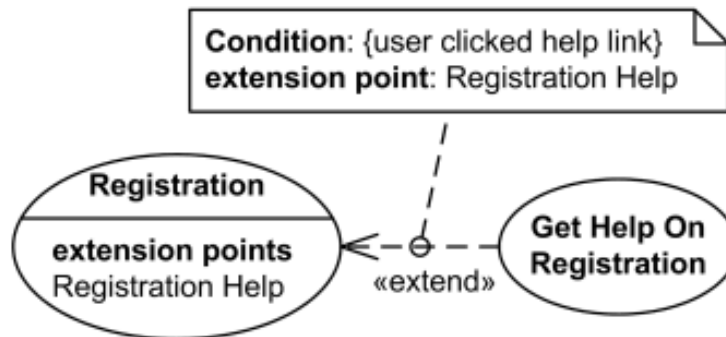
Place extending use cases to the **right** of the base use case.

Use Case with extension points (not tested in exam)

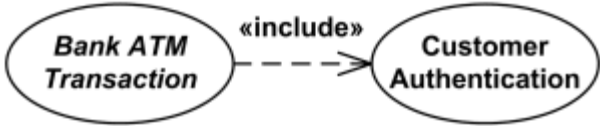
Extension points are detailed version of extend relationships.

Extension points branch off to extending use cases.

The condition of the extend relationship as well as the references to the extension points are optionally shown in a comment note attached to the corresponding extend relationship.

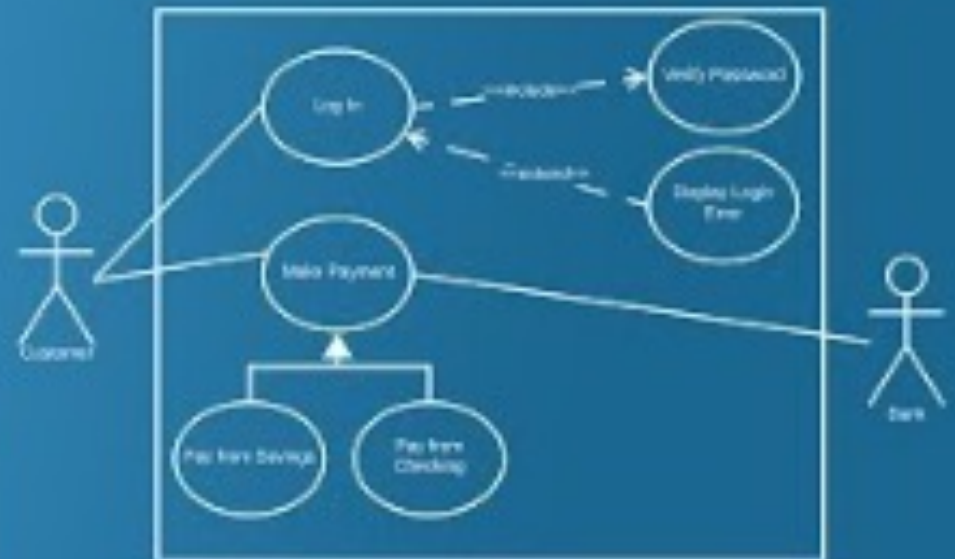


Include VS Extend relationships

Include	Extend
 <pre>graph LR; A([Bank ATM Transaction]) -.-> «include» B([Customer Authentication])</pre>	 <pre>graph LR; A([Bank ATM Transaction]) <.- «extend» B([Help])</pre>
Base use case is <u>incomplete</u> .	Base use case is <u>complete</u> by itself.
Included use case is required, not optional.	Extending use case is optional, supplementary.

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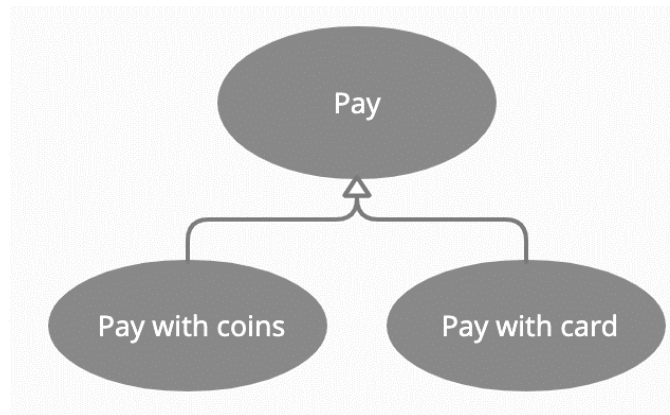
Use Case Diagrams



Generalisation of a Use Case

If you have a coffee machine, you might want to support various payment methods: with coins and cards.

You can create a general use case and two specialised use cases with a generalisation to the general use case.



Generalisation of a Use Case

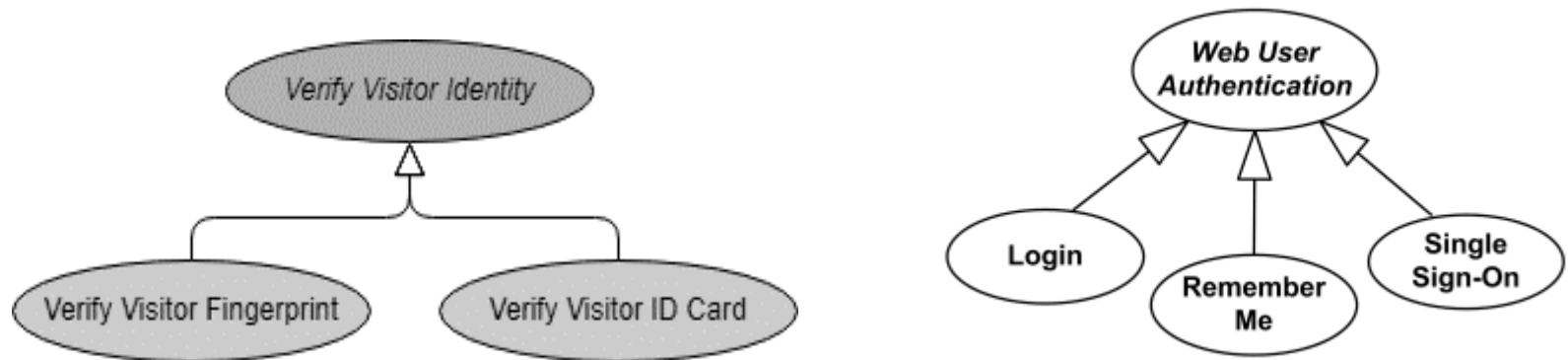
A relationship from a child use case to a parent use case.

This is used when there is a common behavior between two or more use cases and also specialised behavior specific to each use case.

Place specialised use cases below general use case

- This is done to improve the readability of the diagram.

Draw solid directed lines with a large hollow triangle arrowhead from specialised use cases to the general use case.



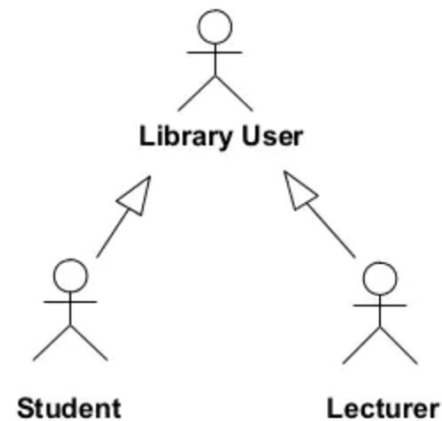
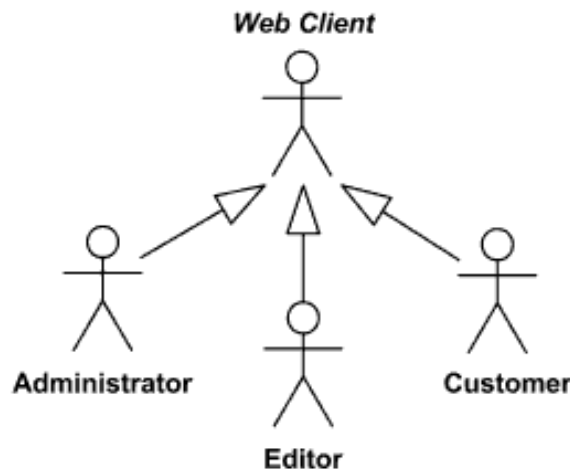
Generalisation of an Actor

A relationship between actors.

Children inherit properties and behaviour of the parent.

Each child can have its unique behaviours.

Draw solid directed lines with a large hollow triangle arrowhead from children to the parent.



Use Case Diagram Guidelines

Naming standards

- Actor name is a noun
- Use case name starts with a verb

Placing of use cases

Usage of system boxes, and

Proper usage of relationships

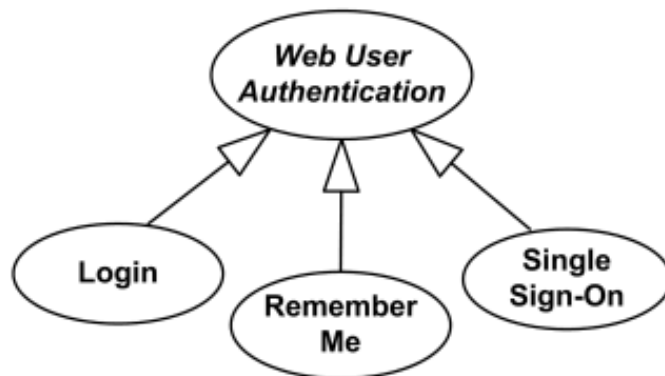
Directions of arrows

Summary

GENERALISATION OF A USE CASE

A relationship from a child use case to a parent use case.

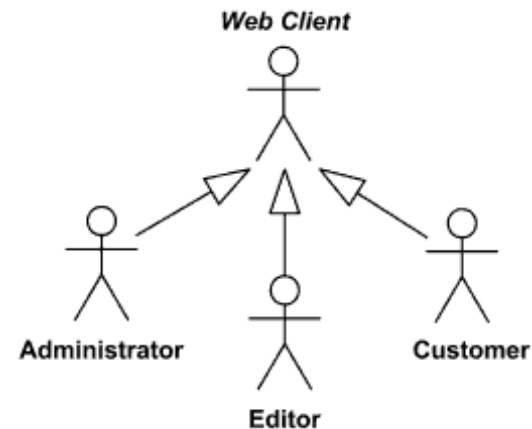
This is used when there is a common behavior between two or more use cases and also specialised behavior specific to each use case.



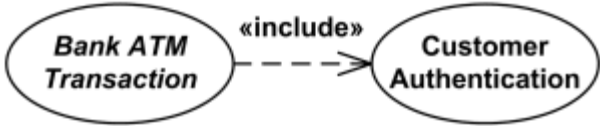
GENERALISATION OF AN ACTOR

A relationship from a child actor to a parent actor.

Children inherit properties and behaviour of the parent.



Summary

Include	Extend
 <pre>graph LR; A([Bank ATM Transaction]) -.-> «include» B([Customer Authentication])</pre> The diagram shows an oval labeled "Bank ATM Transaction" on the left and an oval labeled "Customer Authentication" on the right. A dashed arrow points from the left oval to the right oval, with the label «include» above the arrow.	 <pre>graph LR; A([Bank ATM Transaction]) <--> «extend» B([Help])</pre> The diagram shows an oval labeled "Bank ATM Transaction" on the left and an oval labeled "Help" on the right. A dashed arrow points from the right oval to the left oval, with the label «extend» above the arrow.
Base use case is <u>incomplete</u> .	Base use case is <u>complete</u> by itself.
Included use case is required, not optional.	Extending use case is optional, supplementary.

References:

1. <https://youtu.be/zid-MVo7M-E>
2. <https://creatly.com/blog/diagrams/use-case-diagram-guidelines/>
3. http://theo.cs.ovgu.de/lehre/lehre16s/modelling/01_UseCaseDiagram_slides_2015.pdf
4. <https://t4tutorials.com/common-mistakes-of-use-case-diagrams/>